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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Grade ImprovementExaminationsOctober 2021

Aerospace Engineering

AEROSPACE STRUCTURES

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks

1	1.1	Tail roads contribute to the ________ of the aft portion of the fuselage.	01
	1.2	60 degree banked turn requires a load factor of ________.	01
	1.3	Rough runways will cause the load factor to ________.	01
	1.4	Design Gust Intensity reduces as the velocity ____________.	01
	1.5	Aircraft designed for high altitude flying must be able to withstand ________ loads.	01
	1.6	Increasing the slenderness ratio __________ the critical buckling load.	01
	1.7	Slenderness ratio is the ratio of effective length of column and _______.	01
	1.8	Nature of stresses produces in buckling and bending are ________.	01
	1.9	Keeping loading same but increasing the length, shear stresses in a beam will __________.	01
	1.10	If slenderness ratio=45, which mode of failure will dominate?	01
	1.11	The Miners rule to estimate remaining fatigue life can be calculated as ___________.	01
	1.12	Stress Intensity Factor (K) is not a function of ________ but a function of ________.	01
	1.13	Design life can be written as a multiplication of scatter life and ________.	01
	1.14	The residual strength of a component ________ as crack growth.	01
	1.15	In a safe life based design, the structure must resist load within the __________.	01
	1.16	Use of lap joints is not recommended because _________ are distributed unevenly.	01
	1.17	A rivet is specified as a $20mm$ rivet. What is its shank diameter?	01
	1.18	The centre to centre distance between two consecutive rivets in a row is called ________.	01
	1.19	The efficiency of a riveted joint is defined as the ratio of the ______ to the strength of the unriveted or solid plate.	01
	1.20	Of the three types of structural joints, the fail safe based joint is ___________.	01
2	a	Drawa typical $V-n$ diagram and explain its salient points.	10
	b	Write a short note on Corner Velocity in the V-N diagram with the relative equation.	10
3	a	Elaborate load factor.	10
	b	Describe the effect of the following on wing load. -Fuselage, Nacelle and Wing stores. -Landing and Taxi.	10

4	a	Explain the reason for aero-elastic instabilities with respect to elastic axis and aerodynamic loading.	10
	b	Develop an expression for direct stress for the case of unsymmetrical bending.	10
5	a	Determine the shear flow distribution in the thin walled Z-section.	10
	b	Elaborate shear center in detail.	10
6	a	Differentiate between Elastic Buckling and Inelastic Buckling of columns.	10
	b	Explain Euler's theory of columns stability. Write assumptions and limitations.	10
7	a	Determine the ratio of buckling strengths of two columns, one hollow and the other solid. Both are made of the same material and have the same strength cross sectional area and end conditions. The internal diameter of hollow column is half of its external diameter.	10
	b	An aluminum $(E = 70GPa)$ column built into the ground has length, $L = 2.2m$ and is under axial compressive load P. The dimensions of the cross section are $b = 210mm$ and $d = 280mm$. Calculate the critical load to buckle the column.	10
8	a	Explain 2 Bay Crack Criteria.	10
	b	Write a short note on structural integrity with respect to aircraft structures.	10
9	a	Discuss the life assessment procedures to account for failure modes during aircraft design and qualification process.	10
	b	Narrate in detail Damage Tolerant Design.	10
10	a	Give examples of eccentrically loaded welded joints. How are they analyzed?	10
	b	What are the various permanent and detachable fastenings? Give a complete list with the different types of each category.	10